

Dashboard Screenshot



Problem Design

Background

Superstore is a global retailer selling furniture and office supplies and technology across seven regional markets. Profitability varies widely between these markets. Most of them earn a healthy margin somewhere between about 10 and 27 percent. One market stands apart for the wrong reason. The EMEA market earns a profit margin of only 5.4 percent which is the lowest of every active market in the company.

This is a puzzle rather than a simple case of weak demand. EMEA is not a quiet market. It generates over 800 thousand dollars in sales across 40 countries. The orders are coming in and the customers are clearly there. The sales volume is healthy and only the profit is missing. A market that sells this much yet keeps almost none of the profit is usually a sign that something inside the business is broken rather than something wrong with the customers.

Two competing explanations

There are two very different stories that could explain this and each one leads to a completely different business response.

The first story is economic. EMEA contains many developing and politically unstable countries. Under this view the company is forced to discount heavily across the region just to compete where customers cannot pay full price. If this is true then the correct response is to accept thin margins in EMEA as the cost of operating there.

The second story is operational. Under this view the losses are not forced by the outside world at all. They come from a pricing decision made inside the company in a small number of specific countries. If this is true then the correct response is to fix the pricing rather than accept the loss.

The core question and objective

These two stories cannot both be true and the difference matters because they point to opposite actions. The objective of this dashboard is to determine which explanation is correct. It asks one core question.

Is EMEA losing money because of regional economic hardship or because of a localised pricing failure inside a few specific countries?

Approach

The dashboard combines the internal Superstore sales data with three external World Bank indicators which are wealth and inflation and unemployment. The test is direct. If the economic explanation is correct then the poorer and more unstable countries should carry the heaviest discounts and the deepest losses. If it is wrong then the losses will sit inside a small number of specific countries regardless of their economic health and the cause will be operational. The five visualizations that follow work through this test one factor at a time and then locate the true source of the damage.

Dataset Selection

Internal dataset. Superstore

The Superstore dataset is the retail transaction data explored in Assignment 1. For this report it was filtered down to the EMEA market which contains 5029 transactions across 40 countries and the years 2011 to 2014. The key fields used are Country and Year and Sales and Profit and Discount and Sub-Category.

External datasets. World Bank Open Data

Three indicators were taken from the World Bank for the EMEA countries and the years 2011 to 2014. The World Bank was chosen because it is an internationally recognised and credible source with standardised country level metrics.

1. GDP per Capita PPP. Indicator code NY.GDP.PCAP.PP.CD

This measures the average purchasing power of consumers in each country. It is used to test the idea that poorer countries force heavier discounting.

2. Inflation Consumer Prices annual percent. Indicator code FP.CPI.TOTL.ZG

This measures the rising cost of living. It is used to test whether inflation spikes line up with discounting.

3. Unemployment total percent of labour force modelled ILO estimate. Indicator code SL.UEM.TOTL.ZS

This measures economic stress. It is used to test whether high unemployment lines up with the deep loss margins.

How the external data complements Superstore

The Superstore data shows what is happening inside the business. It shows the sales and the profit and the discount for every order. It does not explain why. The World Bank data adds the economic context for each country and each year. Joining the two on Country and Year lets the dashboard test whether the internal losses are explained by external economic conditions. This is the join that turns a simple sales report into a real business investigation.

Coverage note

Three of the 40 EMEA countries did not have full World Bank data for these years which are Czech Republic and Syria and Yemen. These account for only about 2 percent of the rows and they are dropped from the economic charts. This does not affect the conclusion because none of them are loss making countries.

Citations

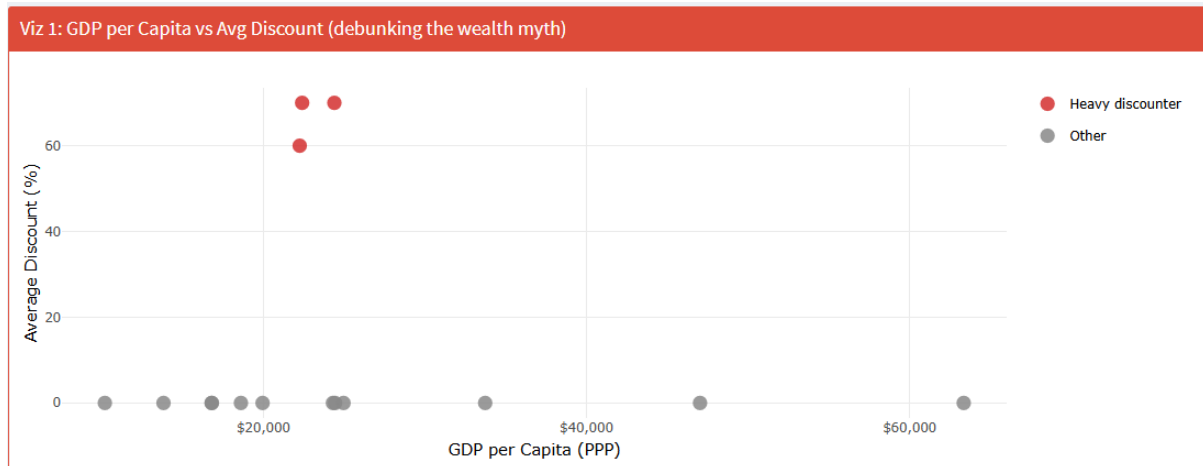
World Bank. (2023). *GDP per capita, PPP (current international \$)* | Data. Worldbank.org. <https://data.worldbank.org/indicator/NY.GDP.PCAP.PP.CD>

World Bank. (2024). *Inflation, consumer prices (annual %)*. Worldbank.org. <https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG>

World Bank. (2025, January 7). *Unemployment, total (% of total labor force) (modeled ILO estimate)*. Worldbank.org. <https://data.worldbank.org/indicator/sl.uem.totl.zs>

Visualization 1. Scatter Plot

GDP per Capita vs Average Discount



What it shows

Each point is one country. The horizontal axis is GDP per capita and the vertical axis is the average discount that Superstore applies there. The three heavy discounters are coloured red.

Interpretation

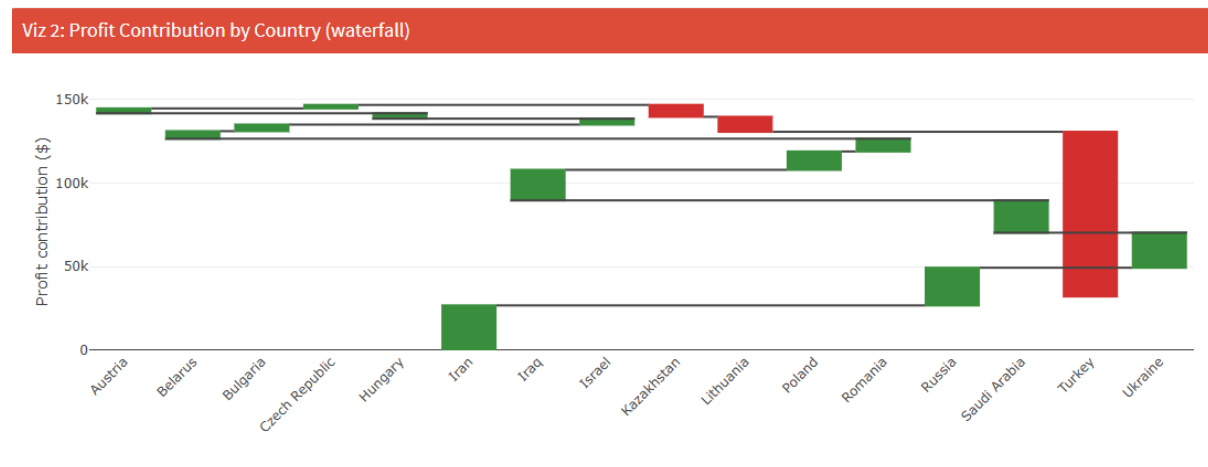
There is no relationship between how wealthy a country is and how hard Superstore discounts it. Thirteen of the sixteen countries sit flat along the zero discount line and they span the full range of wealth from poor to rich. Only three countries discount at all. These are Turkey at 60 percent and Lithuania and Kazakhstan both at 70 percent. These three are not the poorest countries on the chart. They sit at middling wealth levels. The poorest countries such as Ukraine and Iran use no discount at all and stay profitable.

How it answers the problem

This is the chart that breaks the economic hypothesis. If poverty forced discounting then the red points would cluster on the low GDP side. They do not. The discounting is a choice that appears in a few specific countries and it is unrelated to their economic health.

Visualization 2. Waterfall Chart

Profit Contribution by Country



What it shows

This chart builds the EMEA profit total country by country. Green bars add profit and red bars subtract it. The chart is sorted from the strongest contributor to the weakest.

Interpretation

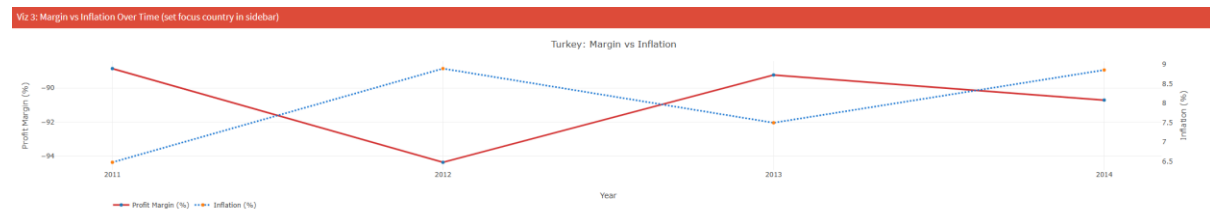
The healthy countries quietly build up a strong regional profit. Iran and Russia and Ukraine each add over twenty thousand dollars. The running total climbs nicely. Then Turkey appears and erases almost all of it with a single drop of about 98 thousand dollars. Turkey alone is responsible for about 86 percent of all the losses in the region. Lithuania and Kazakhstan add smaller red drops. The final regional profit lands at only about 32 thousand dollars which is a fraction of what the healthy countries had built.

How it answers the problem

This chart locates the damage. The problem is not spread across the region. It is concentrated almost entirely in one country. Turkey is the single point of failure that drags the whole EMEA result down to near zero.

Visualization 3. Line Chart

Turkey Margin vs Inflation Over Time



What it shows

This chart follows Turkey across the four years. The red line is Turkey profit margin and the dotted blue line is Turkey inflation rate. The two lines use separate axes.

Interpretation

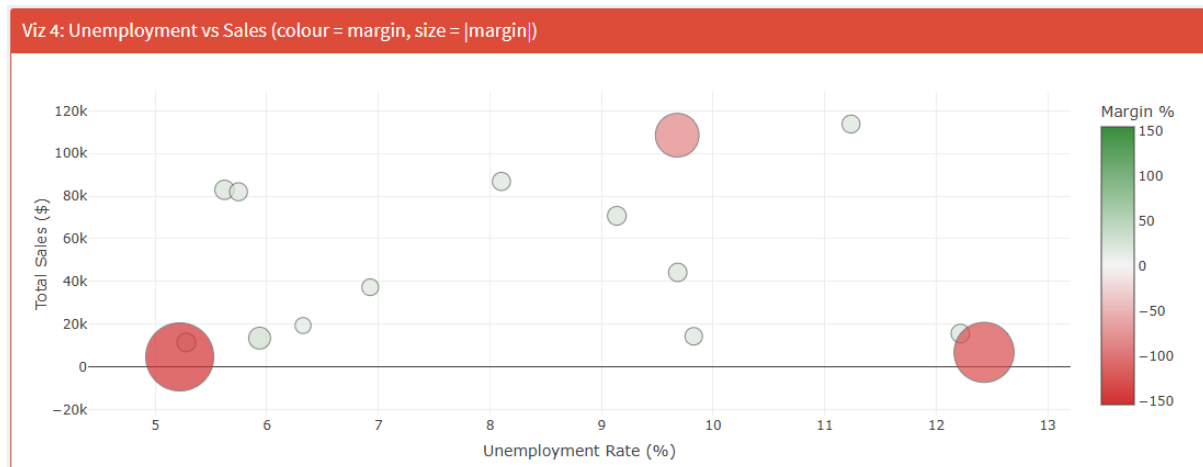
Turkey profit margin is a flat disaster. It sits between minus 89 and minus 94 percent in every single year and it never recovers. Meanwhile inflation moves around between 6.5 and 8.9 percent. The key point is that the two lines do not move together. The margin does not get worse when inflation rises and it does not improve when inflation falls. The margin is simply stuck at a catastrophic level no matter what the economy does.

How it answers the problem

If inflation were the cause of Turkey losses then the margin would track the inflation line. It does not. The loss is constant and independent of the economy. This points to a fixed internal pricing rule rather than a reaction to economic conditions.

Visualization 4. Bubble Chart

Unemployment vs Sales



What it shows

Each bubble is a country. The horizontal axis is unemployment rate and the vertical axis is total sales. The colour shows the profit margin from red for loss to green for profit.

Interpretation

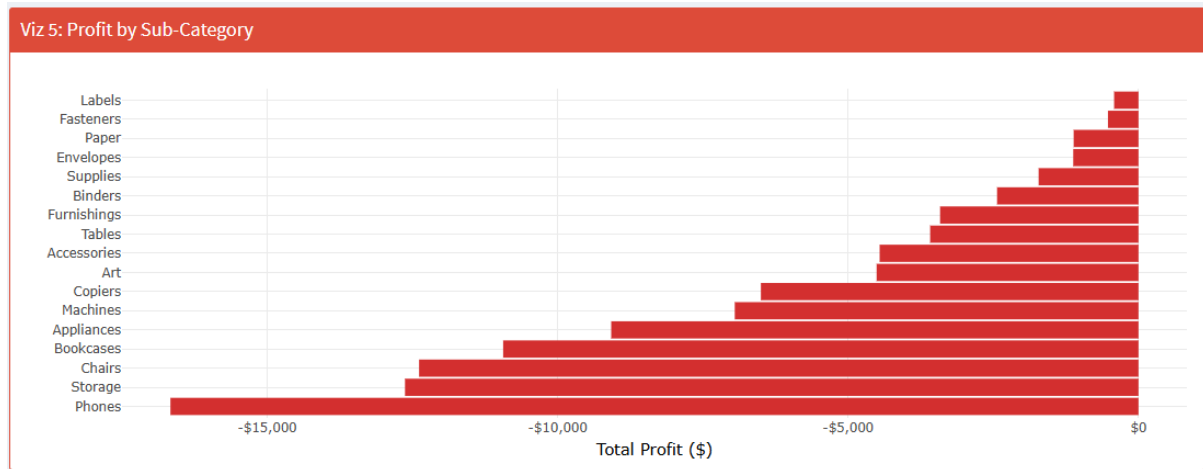
The red loss making bubbles do not sit on the high unemployment side of the chart. Kazakhstan has the worst margin in the region at minus 154 percent yet it has one of the lowest unemployment rates at about 5.2 percent. Turkey sits near the regional median for unemployment. The healthy green countries are spread across both low and high unemployment. So there is no pattern where high unemployment leads to bad margins.

How it answers the problem

This is the third and final economic theory tested and rejected. Unemployment does not explain the losses. A country can have low unemployment and still post the worst margin in the region. The losses are not driven by job market stress.

Visualization 5. Bar Chart

Profit by Sub-Category



What it shows

This horizontal bar chart shows total profit for each product sub-category. Here it is filtered to Turkey only. Green bars are profitable and red bars are losses.

Interpretation

When filtered to Turkey every single sub-category turns red. Every product line loses money and every product line carries the same 60 percent discount. This is the signature of a blanket discount rule rather than a product problem. The largest absolute losses fall on Phones and Storage and Chairs and Bookcases. It is not that a few bad products are dragging Turkey down. The entire catalogue is being sold at a loss because the same heavy discount is applied to everything.

How it answers the problem

This connects the country level failure to the mechanism. Turkey is not losing money because of bad products. It is losing money because a uniform 60 percent discount is applied across the whole range. This is an operational pricing setting and it can be changed.

Data Story

Background

EMEA was the obvious problem market for Superstore. It earned a profit margin of only 5.4 percent which was the lowest of any active market in the company. Yet it was not a small or quiet market. It produced over 800 thousand dollars in sales across 40 countries. Customers were clearly buying. So the question was never about demand. It was about why all that sales volume produced almost no profit. The natural assumption was economic. EMEA holds many developing and unstable economies and the easy explanation is that hard conditions force the company to discount heavily just to compete.

Testing the economic explanation

The dashboard was built to test that assumption rather than accept it. Three economic forces were checked against the data and all three came back negative.

Visualization 1 tested wealth. The scatter of GDP per capita against discount showed no relationship at all. Thirteen of the sixteen countries applied zero discount and they ranged from the poorest to the richest. The poorest countries in the region such as Ukraine and Iran stayed fully profitable. Only three countries discounted and none of them were poor.

Visualization 3 tested inflation. Turkey margin sat at around minus 90 percent in every year from 2011 to 2014 while its inflation rose and fell independently. The two lines never tracked each other. A real cost shock would have moved them together and it did not.

Visualization 4 tested unemployment. The country with the worst margin in the entire region was Kazakhstan at minus 154 percent and it had one of the lowest unemployment rates at about 5.2 percent. High unemployment did not line up with poor margins anywhere on the chart.

Locating the true cause

With every economic explanation ruled out the remaining charts showed where the damage actually came from. Visualization 2 located it geographically. The healthy countries built EMEA profit up steadily and then one country erased almost all of it in a single drop and that country was Turkey. The losses were not spread across a struggling region. They sat inside just three of the sixteen countries which are Turkey and Lithuania and Kazakhstan. All three apply discounts between 60 and 70 percent while every other country discounts nothing. Turkey alone accounts for about 86 percent of the gross losses.

Visualization 5 then revealed the mechanism. Inside Turkey every single product sub-category carries the same 60 percent discount and every one of them loses money. This is not a case of a few weak products. It is a flat discount applied across the entire catalogue. That uniformity is the signature of an operational pricing setting rather than a market reacting to real conditions.

Quantifying the impact

The dashboard makes the cost of this visible with a single filter. With all sixteen countries included the EMEA margin is 4.5 percent. Removing Turkey alone lifts it to 21.3 percent. Removing all three discounters lifts it to 24.4 percent. EMEA is not a weak region at all. Underneath the three discounting countries it is one of the healthiest markets in the company. The poor headline margin was hiding a strong underlying business.

Conclusion and recommendation

The evidence points to one clear conclusion. EMEA losses are not caused by regional economic hardship. They are caused by an aggressive and uniform discount policy in three specific countries and above all in Turkey. This is good news for Superstore because an operational setting is something the company controls directly. The recommended action is not to withdraw from EMEA and not to wait for economies to recover. It is to correct the discount policy in Turkey and Lithuania and Kazakhstan and bring it into line with the zero discount norm that every healthy country in the region already follows. Doing so would turn EMEA from the worst performing market into one of the strongest. The losses were self inflicted and they are fully within the company control to fix.